Light

# GCSE Computer Science – Computational Thinking (Set 3)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

Which statement correctly describes an algorithm?

- ○ **A.** A set of instructions that may or may not eventually complete
- ○ **B.** A finite sequence of well-defined steps that terminates after completing a task
- ○ **C.** Any computer program written in Python or Java
- ○ **D.** A random collection of ideas about solving a problem

Q2 (1 mark)

Flat-pack furniture comes with assembly instructions showing numbered steps with diagrams. Why is this an example of an algorithm?

- ○ **A.** Because it uses pictures instead of words
- ○ **B.** Because it provides a clear sequence of steps that, when followed in order, complete the task of assembling the furniture
- ○ **C.** Because it is printed on paper
- ○ **D.** Because furniture is made of wood or metal

Q3 (1 mark)

What does decomposition mean in computational thinking?

- ○ **A.** Deleting old files from a computer
- ○ **B.** Compressing files to save storage space
- ○ **C.** Breaking down a complex problem into smaller, more manageable sub-problems
- ○ **D.** Converting code from one programming language to another

Q4 (1 mark)

A train journey planner app uses abstraction. What does this mean?

- ○ **A.** The app shows every technical detail about how train engines work
- ○ **B.** The app only works on expensive smartphones
- ○ **C.** The app focuses on relevant information like departure times, platforms, and connections while hiding unnecessary details such as train maintenance schedules

- ○ **D.** The app requires users to have engineering knowledge

Q5 (1 mark)

In a flowchart, what do arrows represent?

- ○ **A.** The direction of flow showing the order in which steps are executed
- ○ **B.** Mathematical calculations
- ○ **C.** User input locations
- ○ **D.** Where errors might occur

Q6 (1 mark)

In AQA pseudocode, what does the statement `score ← score + 10` do?

- ○ **A.** Checks if score equals 10
- ○ **B.** Creates a new variable called 10
- ○ **C.** Displays the value 10 on screen
- ○ **D.** Adds 10 to the current value of score and stores the result back in score

Q7 (1 mark)

In the algorithm step 'total ← price × quantity', what type of operation is this?

- ○ **A.** Processing - a calculation is being performed and the result stored
- ○ **B.** Output - data is being displayed
- ○ **C.** Input - data is being received
- ○ **D.** Decision - a choice is being made

Q8 (1 mark)

Where does linear search begin looking for an item in a list?

- ○ **A.** At the middle of the list
- ○ **B.** At the end of the list
- ○ **C.** At a random position
- ○ **D.** At the first item and proceeds sequentially through each item

Q9 (1 mark)

What must be true about data before binary search can be used?

- ○ **A.** The data must already be sorted in order
- ○ **B.** The data must be stored in a database
- ○ **C.** The data must contain only numbers
- ○ **D.** The data must have fewer than 100 items

Q10 (1 mark)

During bubble sort, when are two adjacent items swapped?

- ○ **A.** They are always swapped regardless of their values
- ○ **B.** They are swapped when they are in the wrong order relative to each other
- ○ **C.** They are swapped only if both values are negative
- ○ **D.** They are never swapped - bubble sort uses a different mechanism

Q11 (1 mark)

In merge sort, why does the algorithm keep dividing until sublists contain only one element?

- ○ **A.** Because single elements use less memory
- ○ **B.** Because single elements can be deleted more easily
- ○ **C.** Because the algorithm crashes with larger sublists
- ○ **D.** Because a single element is automatically sorted - there is nothing to compare it with

Q12 (1 mark)

When comparing algorithm efficiency at GCSE level, what aspect is primarily considered?

- ○ **A.** How much money the algorithm costs to develop
- ○ **B.** How visually appealing the code looks

- ○ **C.** How much time the algorithm takes to complete its task
- ○ **D.** How many programmers are needed to write it

Q13 (2 marks)

Consider the algorithm:

```
total ← 0
FOR num ← 2 TO 4
   total ← total + (num × 2)
ENDFOR
What is the final value of total?
```

- ○ **A.** 18
- ○ **B.** 12
- ○ **C.** 24
- ○ **D.** 8

Q14 (2 marks)

A sorted list contains 128 items. What is the maximum number of comparisons binary search needs to find any item (or confirm it is not present)?

- ○ **A.** 128 comparisons
- ○ **B.** 7 comparisons
- ○ **C.** 64 comparisons

- ○ **D.** 8 comparisons

Q15 (2 marks)

The list [7, 3, 9, 2] is being sorted using bubble sort. After the first complete pass, what does the list look like?

- ○ **A.** [2, 3, 7, 9]
- ○ **B.** [7, 3, 2, 9]
- ○ **C.** [3, 7, 2, 9]
- ○ **D.** [3, 7, 9, 2]

Q16 (2 marks)

The list [4, 8, 2, 6] is being sorted using merge sort. What sublists exist after the division phase is complete?

- ○ **A.** [[4, 8], [2, 6]]
- ○ **B.** [[2, 4, 6, 8]]
- ○ **C.** [[4], [8], [2], [6]]
- ○ **D.** [[4, 2], [8, 6]]

Q17 (2 marks)

A music streaming service allows users to search their personal playlists. Users frequently add and remove songs, so playlists are unsorted. Which search algorithm is more practical and why?

- ○ **A.** Binary search because it is always faster than linear search
- ○ **B.** Linear search because it works on unsorted data without requiring the playlist to be sorted first
- ○ **C.** Neither algorithm would work for music data
- ○ **D.** Both algorithms would sort the playlist automatically

Q18 (2 marks)

A team is developing a fitness tracking app. Using decomposition, which breakdown of the project makes sense?

- ○ **A.** Create everything as one massive function with no separation
- ○ **B.** Separate into: activity tracking (steps, distance, calories), workout logging (exercises, sets, reps), progress visualisation (charts, statistics), and social features (friends, challenges, sharing)
- ○ **C.** Only focus on step counting and ignore all other features
- ○ **D.** Start with social features and never add fitness tracking

Q19 (2 marks)

A video game simulates football matches. Which use of abstraction is appropriate?

- ○ **A.** Modelling the exact chemical composition of the grass on the pitch
- ○ **B.** Requiring players to understand advanced physics equations to play
- ○ **C.** Only showing the final score with no match simulation

○ **D.** Tracking player positions, ball physics, and team formations while ignoring details like individual blade movements or exact stitching patterns on the ball

Q20 (2 marks)

Two sorting algorithms are tested on the same computer with the same data. Algorithm P takes 3 seconds; Algorithm Q takes 0.3 seconds. What can be concluded?

○ **A.** Algorithm Q uses more memory

○ **B.** Algorithm P produces better sorted results

○ **C.** Algorithm Q is more time-efficient for this data on this computer

○ **D.** Nothing can be concluded from timing data

Q21 (2 marks)

A flowchart has: Start → Input age → Diamond 'age >= 18?' with 'Yes' going to 'OUTPUT "Adult"' and 'No' going to 'OUTPUT "Minor"' → both paths lead to End. What outputs when age is 18?

○ **A.** Adult

○ **B.** Minor

○ **C.** Both Adult and Minor

○ **D.** Nothing is output

Q22 (2 marks)

Consider the algorithm:

```
x ← 5
y ← 3
x ← x + y
y ← x - y
OUTPUT y
What value is output?
```

- ○ **A.** 3
- ○ **B.** 2
- ○ **C.** 8
- ○ **D.** 5

Q23 (2 marks)

An algorithm is written in pseudocode, then converted to a flowchart, then implemented in Python. How do these three versions relate?

- ○ **A.** They are three completely different algorithms
- ○ **B.** The flowchart version will run faster than the others
- ○ **C.** Only the Python version is a real algorithm
- ○ **D.** They are three representations of the same algorithm - the underlying logic remains constant

Q24 (3 marks)

A sorted list contains [15, 25, 35, 45, 55, 65, 75, 85]. Binary search is used to find value 75. Show the sequence of comparisons.

- ○ **A.** 15, 25, 35, 45, 55, 65, 75 - checking from the start

- **B.** 45 (middle), then 65 (middle of right half), then 75 - found in 3 comparisons
- **C.** 85, 75 - checking from the end
- **D.** All 8 elements are checked simultaneously

Q25 (3 marks)

The list [4, 1, 3, 2] is sorted using bubble sort. How many passes are required to fully sort the list, and what does it look like after each pass?

- **A.** 3 passes: Pass 1 → [1, 3, 2, 4], Pass 2 → [1, 2, 3, 4], Pass 3 (verification) → [1, 2, 3, 4]
- **B.** 2 passes: Pass 1 → [1, 3, 2, 4], Pass 2 → [1, 2, 3, 4]
- **C.** 1 pass: [1, 2, 3, 4]
- **D.** 4 passes required regardless of data

Q26 (3 marks)

Trace the complete merge sort process for the list [6, 2, 8, 4].

- **A.** Compare adjacent pairs until sorted: [2, 6, 4, 8] → [2, 4, 6, 8]
- **B.** Divide: [6, 2, 8, 4] → [6, 2] and [8, 4] → [6], [2], [8], [4]. Merge: [2, 6], [4, 8]. Final merge: [2, 4, 6, 8]
- **C.** Find minimum repeatedly: 2, then 4, then 6, then 8
- **D.** Randomly shuffle until sorted

Q27 (3 marks)

An e-commerce warehouse has 500,000 products stored in a database sorted by product code. Staff need to quickly check if specific products are in stock. Which search algorithm should be used and why?

- ○ **A.** Linear search because it is easier to implement
- ○ **B.** No search algorithm can handle 500,000 items
- ○ **C.** Binary search because the sorted database allows each comparison to eliminate half the remaining products, finding any item in approximately 19 comparisons instead of potentially 500,000
- ○ **D.** Products should be searched by colour instead of code

Q28 (3 marks)

A social media platform needs to sort users' news feeds. New posts arrive constantly, and feeds must be re-sorted frequently to show newest content first. Given that data changes very often and re-sorting happens continuously, which approach makes most sense?

- ○ **A.** Use bubble sort because it can efficiently handle nearly-sorted data when new items are added to an almost-sorted list
- ○ **B.** Use merge sort for every re-sort regardless of how much the data has changed
- ○ **C.** Never sort the feed - show posts randomly
- ○ **D.** Only sort the feed once per year

Q29 (3 marks)

A company is developing a smart home system. How should computational thinking principles be applied?

- ○ **A.** Build everything as one interconnected piece with no modular structure
- ○ **B.** Only implement security features and ignore everything else
- ○ **C.** Decomposition: separate into modules for lighting control, heating/cooling, security (cameras/alarms), and entertainment systems. Abstraction: each module focuses on relevant data (light levels, temperatures, motion detection, audio/video sources) while hiding implementation details from users and other modules
- ○ **D.** Require users to have electrical engineering degrees to operate the system

Q30 (3 marks)

A developer needs to create a hotel booking system. They have never built a reservation system before. What systematic approach should they follow?

- ○ **A.** Start typing code immediately without any design or planning
- ○ **B.** Copy another hotel's system without understanding how it works
- ○ **C.** Declare the task impossible and give up
- ○ **D.** Apply computational thinking: decompose into sub-problems (room inventory management, date availability checking, booking processing, payment handling, confirmation generation); abstract by focusing on essential data (room types, dates, prices, guest details) while ignoring irrelevant details (room paint colours, furniture brands); design algorithms for each sub-problem before implementing

Total: **55** marks • Time: **30 min**

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